

MTF-JEPA-MAE-VICReg

A Representation Encoder for World Models

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Abstract—MTF-JEPA-MAE-VICReg is a C++20/LibTorch encoder that learns numerical representations of sensor histories without manually labeled targets. It combines temporal statistics and frequency information, handles missing measurements, and produces vectors for each history and sensor channel. The core encoder and CPU/CUDA training pipeline are implemented and working. We seek support to refine and validate this foundation as an observation component for robotic world models.

Index Terms—Robotics, world models, representation learning, sensor histories.

I. THE ROBOTICS OPPORTUNITY

A robotic world model predicts the effects of actions. Its observation encoder turns sensor histories into embeddings: learned numerical vectors used by a predictor. Inputs could include joint motion, inertial measurements, motor current, and force readings. This could complement a separate vision encoder.

II. REPRESENTATION AND LEARNING

The input X is a rank-four real-valued array with shape $[B, C, H, F]$:

- B histories processed together (batch size)
- C sensor streams or groups (channels)
- H time steps in each history
- F measurements per channel at each time step

A Boolean mask of the same shape marks each measurement as observed or missing.

MTF (multiscale time-frequency) analysis summarizes windows of several lengths using means, standard deviations, and oscillation strengths. These summaries become tokens: learned vectors tagged by channel, window length, time/frequency type, and position. The shared encoder mixes information across valid tokens. Averaging its outputs yields a vector per history and per channel; flags identify outputs with no observations.

Learning combines four objectives. JEPA (joint-embedding predictive architecture) predicts hidden token embeddings from visible context [1]. Targets come from a teacher: a slowly averaged copy of the token builder and encoder. MAE (masked autoencoder) reconstructs window summaries [2]. Time-frequency alignment encourages agreement between the two views. VICReg (variance-invariance-covariance regularization) aligns lightly perturbed views while preserving variation across examples and reducing redundant vector dimensions [3].

III. DEVELOPMENT TOWARD A WORLD MODEL

The proposed next step learns how embeddings change under robot actions:

$$\hat{z}_{t+1} = f_{\phi}(z_t, a_t). \quad (1)$$

Here t is a time step, z_t the current history’s embedding, a_t the action, f_{ϕ} a predictor with learned parameters ϕ , and $\hat{z}_{t+1}$ the predicted next embedding. Current training reconstructs information within an observed history; future prediction and control integration remain next steps.

A manipulation-task pilot would compare learned and hand-designed baselines on unseen robot trajectories, measuring prediction accuracy, missing-sensor robustness, inference time, and control utility. Training and evaluation histories must be temporally separated to avoid overlap.

IV. READINESS AND COMPLETION

The implementation supports CPU/CUDA training, saving and resuming training state, and embedding export in a Debian environment. Tests cover missing data, teacher updates, and numerical consistency. Robotics performance and real-time suitability remain to be measured.

This implemented core demonstrates our capability to build and test self-supervised representation systems. We seek support to refine it through completion as a robotic world-model component, including validation on real robot trajectories and downstream tasks.

V. SOURCE CODE

The repository contains the standalone encoder, training workflow, and setup instructions for review and further refinement.

https://github.com/savethebeesandseeds/cuwacunu_embedding

REFERENCES

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